

# **Morningstar Sustainability Ratings on Mutual Funds Performance**

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## **1. Introduction:**

The Morningstar Sustainability Rating is a mechanism intended to assist investors in assessing the efficiency with which firms in mutual funds are handling their Environmental Social Governance (ESG) risks and opportunities. This report examines whether these ratings affect mutual funds performance, specifically their impact on their gross and net returns. By applying December 2018 data on UK equity mutual funds, we examine the impact of sustainability ratings on their net and gross returns. Our analysis focuses on the following two core questions: whether sustainability ratings influence a fund's gross and net returns, and if higher ratings are being linked with greater returns.

To answer these questions, we use multiple regression analysis to assess the relationship between sustainability ratings and mutual fund performance. The analysis process includes inspection of the given dataset, incorporating statistical methods (Ordinary Least Squares) to estimate a relationship of these ratings on fund performance, and building regression models. The results of this research is presented to our fund manager.

## **2. Dataset Analysis:**

### **2. 1: Dataset Overview**

The dataset consists of a cross-section of 899 UK equity mutual fund companies as of December 2018. Each firm is represented by 9 variables (4 quantitative and 5 qualitative variables [Table. 1]).

**Table. 1: Variable Features & Descriptions**

| <b>Variables</b>     | <b>Units</b>                                      | <b>Descriptions (Type of variable)</b>                                                                    |
|----------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Name                 | -                                                 | Name of funds (Categorical variable)                                                                      |
| Fundid               | -                                                 | ID of funds (Categorical variable)                                                                        |
| Globalcategory       | -                                                 | Style of funds (Categorical variable)                                                                     |
| Size                 | Millions of pounds                                | Size of funds (Quantitative-continuous variable)                                                          |
| Investmentarea       | -                                                 | Region where the fund invests (Categorical Variables)                                                     |
| Sustainabilityrating | Low, Below Average, Average, Above Average & High | Sustainability Ratings of the funds (Categorical Variables)                                               |
| Fund_age             | Years                                             | Indicates the age of the mutual fund business (Quantitative-continuous variable)                          |
| Gross_return         | Percentage                                        | Gross value of returns- obtained before deduction of fees (Quantitative-continuous variable)              |
| Net_return           | Percentage                                        | Net value of returns of mutual funds- obtained after deduction of fees (Quantitative-continuous variable) |

Sustainability ratings were categorized on 5 different values ranging from [Low to High]

Summary statistics of the given quantitative set of variables are as follows [Table. 2]:

**Table. 2: Key Variable Statistics**

| <b>Variable</b>           | <b>No. of observations</b> | <b>Mean</b> | <b>Std Dev</b> | <b>Min</b> | <b>Max</b> |
|---------------------------|----------------------------|-------------|----------------|------------|------------|
| Size                      | 899                        | 486.608     | 1039.74        | 0.000948   | 15772.43   |
| Fund_age                  | 899                        | 16.20792    | 12.72884       | 0.0833333  | 81.91666   |
| Net_return                | 899                        | -5.409649   | 2.523065       | -14.4619   | 11.96832   |
| Gross_return              | 891                        | -5.337448   | 2.534757       | -14.41685  | 12.12578   |
| Sustainability<br>Ratings | 0                          | -           | -              | -          | -          |

Size: The average value of each mutual fund business is £486.6 million, with a standard deviation of £1,039.74, indicating substantial variability and an appropriately broad distribution range. The minimum and maximum mutual fund sizes in the sample are £948 and £15.7 billion, respectively. This illustrates a significant disparity in fund values.

Fund\_age: The average age of the mutual funds in the sample is about 16 years, as indicated by the mean of 16.2079 years, along with a standard deviation of 12.72884 years. The youngest fund is about 1 month old, and the oldest firm has been in business for almost 82 years.

Gross\_return: Refers to the total return earned by an investment before deducting any expenses, fees, or taxes, with a mean of -5.33744% and a standard deviation of 2.534757%. There are 8 missing observations, as some firms didn't disclose this information in the dataset. The fund with the poorest gross return lost 14.416% from the total value. The highest gross return of the firms in the sample was 12.12%. Gross returns are useful for evaluating a fund manager's skill and comparing performance across funds.

Net return: Denotes the net profit or loss resulting from an investment after deducting all expenditures, fees, and taxes. Most firms in the sample perform poorly, as shown by the fund's average net return rate's mean of -5.49064% and standard deviation of 2.523065%, showing high volatility. In December 2018, the worst-performing UK Equity Mutual Fund lost 14.461% of its assets, while the best-performing fund gained 11.96%. Net returns show profitability and help investors understand their returns.

## 2.2: Observations:

- I. Nearly every business [Fig. 1] in the study had net returns more negative than gross returns, and the shift was modest, indicating the effect of tax and fees on returns.

|   | name                                    | sustainabilit... | size     | net_return | gross_re... | fund_age |
|---|-----------------------------------------|------------------|----------|------------|-------------|----------|
| 1 | Vanguard LifeStrategy 100% Equity A Acc | Below Aver...    | 1067.282 | -6.39      | -6.37       | 7.5      |
| 2 | Fidelity Sustainable Asia Eq A Acc      | Average          | 87.82265 | -2.896608  | -2.811358   | 7.416667 |
| 3 | WS Saracen Global Inc and Growth BAcc   | Above Ave...     | 112.0618 | -6.583075  | -6.5        | 7.5      |
| 4 | Fidelity China W Acc                    | Average          | 121.3379 | -2.902691  | -2.820153   | 7.25     |
| 5 | Aviva Investors US Eq Inc 3 GBP Acc     | Low              | 158.5993 | -9.5413    | -9.471134   | 7.416667 |
| 6 | MI Metropolis Value S Acc               | Below Aver...    | 130.6352 | -7.512521  | -7.434174   | 7.666667 |

**Fig. 1: Sample browse of data**

- II. From the dataset, most net and gross rates were negative; few firms had positive returns.
- III. Notably, fund size and age ranges were too large (Table. 2). So, we checked the percentile distributions of these parameters [Fig. 2].

| Stats | size     | fund_age |
|-------|----------|----------|
| Mean  | 486.608  | 16.20792 |
| SD    | 1039.74  | 12.72884 |
| Min   | .000948  | .0833333 |
| Max   | 15772.43 | 81.91666 |
| p1    | 1.294697 | .3333333 |
| p5    | 7.450045 | 1.166667 |
| p50   | 159.1595 | 13       |
| p75   | 436.6542 | 23.91667 |
| p99   | 5492.017 | 51.25    |

**Fig. 2: Stats for size & age**

### **2.3: Correlations:**

Correlations analysis helps to understand the relationship between variables.

**Table. 3: Correlations between Variables**

|                     | <b>Gross_return</b> | <b>Net_return</b> | <b>Size</b> | <b>Fund_age</b> |
|---------------------|---------------------|-------------------|-------------|-----------------|
| <b>Gross_return</b> | 1.0000              |                   |             |                 |
| <b>Net_return</b>   | 0.9999              | 1.0000            |             |                 |
| <b>Size</b>         | -0.0225             | -0.0197           | 1.0000      |                 |
| <b>Fund_age</b>     | -0.0408             | -0.0446           | 0.0518      | 1.0000          |

Gross\_return and net\_return are positively correlated at 0.9999. This connection suggests comparable values as 1 is the strongest correlation.

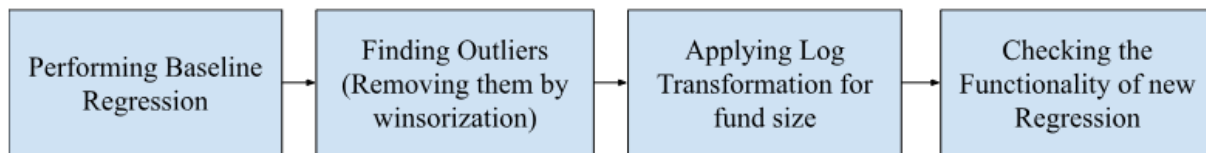
Gross return is negatively correlated with size of -0.0225. Creating an inverse linear relationship between fund size and gross return. Also, net\_return and size are -0.0197. Additionally, this connection has the lowest matrix correlation [Table. 3].

Size and fund\_age correlation is 0.0518, a moderate positive correlation of 5%.

Except for the nearly perfect positive correlation between gross return and net income, variables are poorly correlated. The dataset's categorical sustainability ratings prohibit linkages with other parameters, emphasising the need to integrate them.

### **3. Methodology:**

- A. To assess the influence of sustainability ratings on returns, it is necessary to construct a regression model with sustainability ratings as the explanatory variable, while gross and net returns serve as the explained variables. These two regressands will be analysed separately in distinct regressions, as their high collinearity would obscure their individual contributions if included in a single regression equation, potentially leading to various complications in the analysis.
- B. The aforementioned dataset possesses sufficient parameters for regression analysis. Given that the size and age of mutual funds are significant based on prior research, we include them as regressors in our model. To integrate ordinal information, such as sustainability ratings, it is beneficial to utilise dummy variables for these ratings. The subsequent analysis will employ a multiple regression, utilising four dummy variables for sustainability ratings, two regressors for fund characteristics (size and age), along with an intercept and an error term to account for unobserved components in the regression equation.
- C. The base case regression in our technique is merely the first block, but we induce alterations [Fig. 3] to improve multiple regression. It is unique and employs our sensible regression modelling methods to create a final model. However, it cannot be asserted that it is the optimal model; the final model will serve as our framework for estimating performance ratings.



**Fig. 3: Block diagram of our analysis**

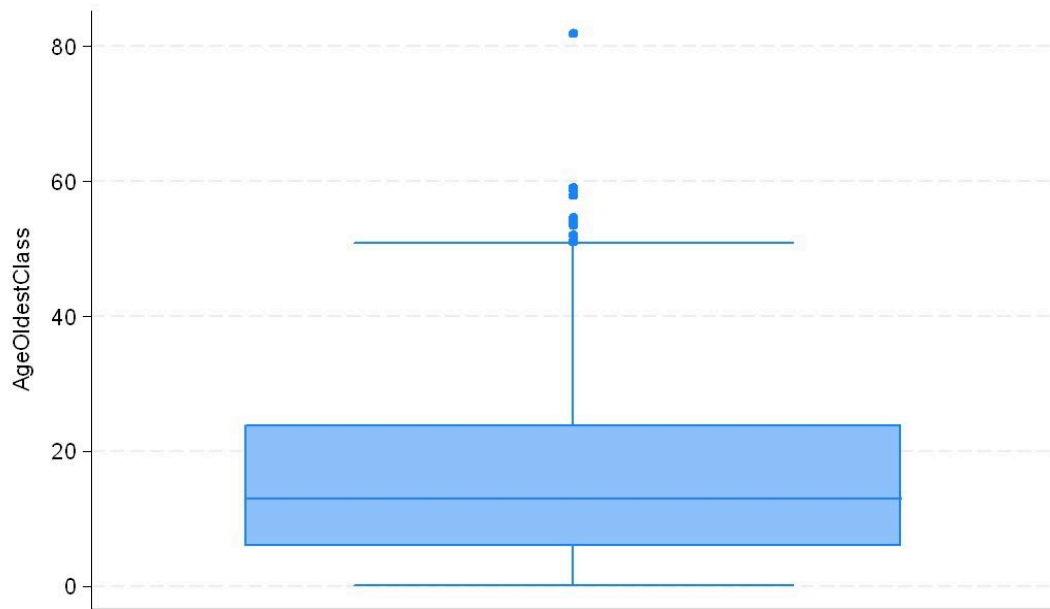
#### **3.1.1: Baseline regressions:**

$$net\_return = \beta_0 + \beta_1 B\_AVG + \beta_2 AVG + \beta_3 A\_AVG + \beta_4 HIGH + \beta_5 size + \beta_6 fund\_age + u$$

$$gross\_return = \beta_0 + \beta_1 B\_AVG + \beta_2 AVG + \beta_3 A\_AVG + \beta_4 HIGH + \beta_5 size + \beta_6 fund\_age + u$$

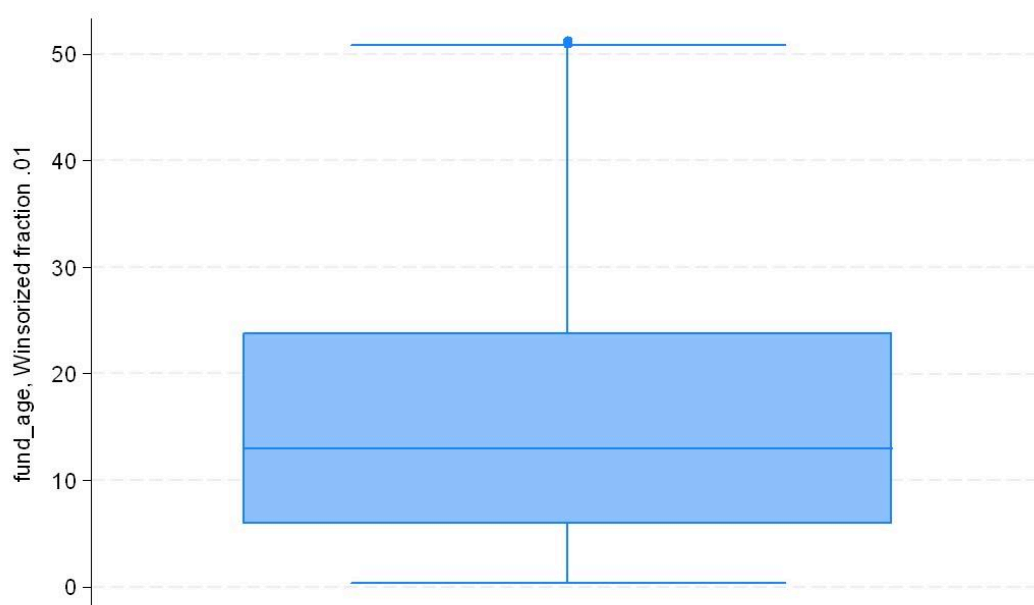
There are only 4 dummy variables to avoid the issue of dummy variable trap, where the  $\beta_0$  = intercept dummy represents the “low” category of funds,  $u$ =error term.

### 3.1.2: Finding Outliers:

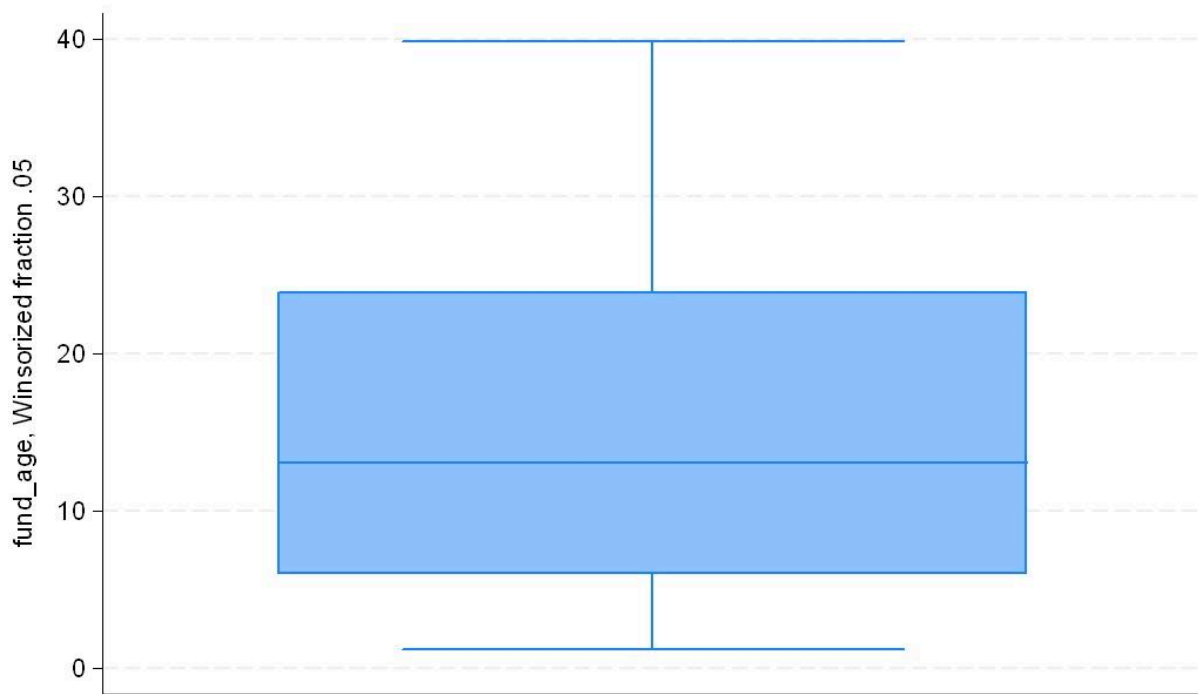


**Fig. 4: Depicts outliers of fundage beyond 60 years**

Winsorization: To decrease false outliers, winsorizing limits extreme values in statistical data.



**Fig. 5: Depicts a single outlier of fundage at 50 years after winsorization with [1, 99] percentile cutoff.**



**Fig. 6: Winsorized model with [5,95] cutoff has no outliers.**

Since both models [Fig. 5, Fig. 6] are very similar, its confusing to select them for our regression; hence, we include both of them in our model estimation.

### **3.1.3: Log transformation of fundsize:**

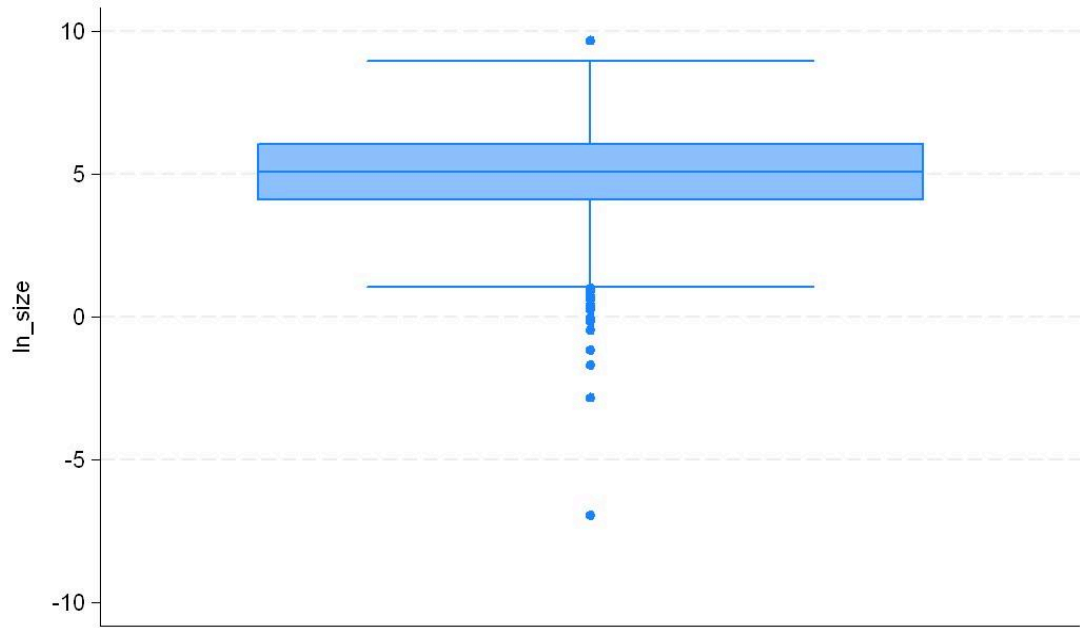
As the range of size is too large, we perform log transformation to cut down its range [Fig. 7].

| Stats | size            | ln_size          |
|-------|-----------------|------------------|
| Mean  | <b>486.608</b>  | <b>4.990513</b>  |
| SD    | <b>1039.74</b>  | <b>1.722472</b>  |
| Min   | <b>.000948</b>  | <b>-6.961156</b> |
| Max   | <b>15772.43</b> | <b>9.666018</b>  |
| p1    | <b>1.294697</b> | <b>.2582767</b>  |
| p5    | <b>7.450045</b> | <b>2.00822</b>   |
| p50   | <b>159.1595</b> | <b>5.069907</b>  |
| p75   | <b>436.6542</b> | <b>6.079142</b>  |
| p99   | <b>5492.017</b> | <b>8.611051</b>  |

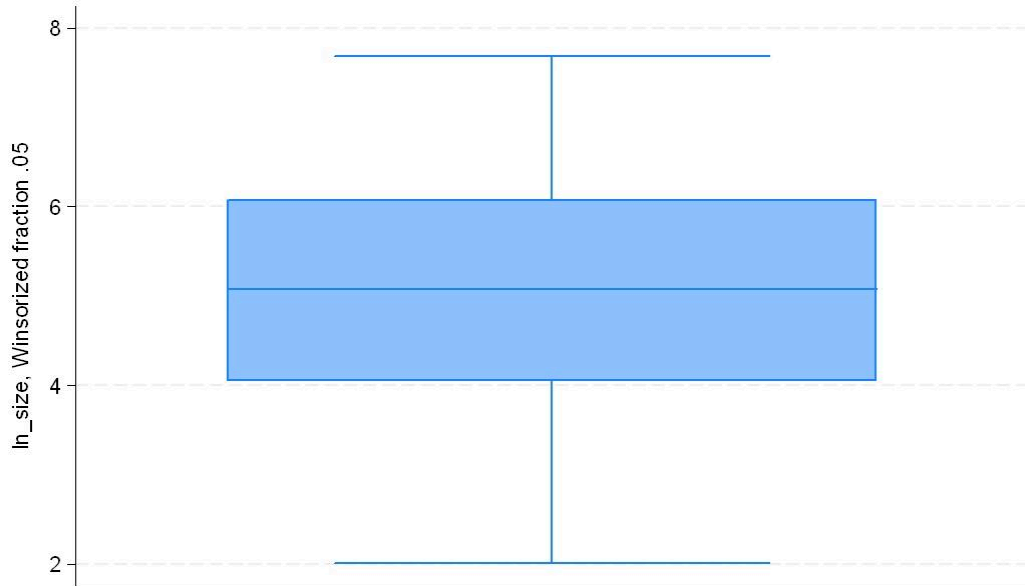
**Fig. 7: size ln\_size stats**

Due to the presence of outliers again in ln\_size, we winsorize them at 5% [Fig. 9].





**Fig. 8: Pre-winsorization of  $\ln\_size$**



**Fig. 9: Post-winsorization of  $\ln\_size$**

Incorporating the transformations of explanatory variables, we obtain the below estimations for fund performance:

Regression (i) -

$$net\_return = \beta_0 + \beta_1 B\_AVG + \beta_2 AVG + \beta_3 A\_AVG + \beta_4 HIGH + \beta_5 wins\_ln\_size + \beta_6 wins\_fund\_age\_one + u$$

Regression (ii)-

$$net\_return = \beta_0 + \beta_1 B\_AVG + \beta_2 AVG + \beta_3 A\_AVG + \beta_4 HIGH + \beta_5 wins\_ln\_size + \beta_6 wins\_fund\_age\_five + u$$

Regression (iii)-

$$gross\_return = \beta_0 + \beta_1 B\_AVG + \beta_2 AVG + \beta_3 A\_AVG + \beta_4 HIGH + \beta_5 wins\_ln\_size + \beta_6 wins\_fund\_age\_one + u$$

Regression (iv)-

$$gross\_return = \beta_0 + \beta_1 B\_AVG + \beta_2 AVG + \beta_3 A\_AVG + \beta_4 HIGH + \beta_5 wins\_ln\_size + \beta_6 wins\_fund\_age\_five + u$$

### 3.1.4: Functionality check of the new regression:

The Ramsey RESET test serves as a diagnostic instrument in regression analysis, aimed at detecting model misspecification by the identification of omitted variables, incorrect functional forms (log transformations), or interaction effects, thereby ensuring the regression model accurately reflects the data.

H0: the model is correctly specified

H1: the model is incorrectly specified

```
. reg net_return B_AVG AVG A_AVG HIGH wins_ln_size_ wins_fund_age_one
```

| Source   | SS         | df  | MS         | Number of obs | = | 899    |
|----------|------------|-----|------------|---------------|---|--------|
| Model    | 332.523887 | 6   | 55.4206479 | F(6, 892)     | = | 9.18   |
| Residual | 5384.01521 | 892 | 6.0358915  | Prob > F      | = | 0.0000 |
|          |            |     |            | R-squared     | = | 0.0582 |
|          |            |     |            | Adj R-squared | = | 0.0518 |
| Total    | 5716.5391  | 898 | 6.36585646 | Root MSE      | = | 2.4568 |

| net_return        | Coefficient | Std. err. | t      | P> t  | [95% conf. interval] |           |
|-------------------|-------------|-----------|--------|-------|----------------------|-----------|
| B_AVG             | .4351299    | .3316761  | 1.31   | 0.190 | -.2158267            | 1.086086  |
| AVG               | .9177086    | .308528   | 2.97   | 0.003 | .3121832             | 1.523234  |
| A_AVG             | 1.115522    | .3258554  | 3.42   | 0.001 | .4759898             | 1.755055  |
| HIGH              | 2.534668    | .3950758  | 6.42   | 0.000 | 1.759281             | 3.310054  |
| wins_ln_size_     | -.0627303   | .0570638  | -1.10  | 0.272 | -.1747252            | .0492647  |
| wins_fund_age_one | -.0098462   | .0068281  | -1.44  | 0.150 | -.0232472            | .0035548  |
| _cons             | -5.860735   | .3827061  | -15.31 | 0.000 | -6.611845            | -5.109626 |

```
. estat ovtest
```

Ramsey RESET test for omitted variables  
Omitted: Powers of fitted values of **net\_return**

H0: Model has no omitted variables

F(3, 889) = 0.96  
Prob > F = 0.4133

Fig. 10: Sample output of our Ramsey test on Regression-(i)

**Table. 4: Ramsey Reset test outputs**

| <b>No.</b> | <b>Test: Ramsey's Reset Test</b>          | <b>p-value</b> | <b>Conclusion</b>                    |
|------------|-------------------------------------------|----------------|--------------------------------------|
| 1.         | <i>Baseline regression: net_returns</i>   | p=0.4023       | H0: The model is correctly specified |
| 2.         | <i>Baseline regression: gross_returns</i> | p=0.3670       | H0: The model is correctly specified |
| 3.         | <i>Our regression-i</i>                   | p=0.4133       | H0: The model is correctly specified |
| 4.         | <i>Our regression-ii</i>                  | p=0.3779       | H0: The model is correctly specified |
| 5.         | <i>Our regression-iii</i>                 | p=0.3998       | H0: The model is correctly specified |
| 6.         | <i>Our regression-iv</i>                  | p=0.3529       | H0: The model is correctly specified |

The p-values are all above typical significance levels (e.g., 0.05), so the null hypothesis (H0: the model is correctly specified) is not rejected for any model. This indicates no evidence of misspecification in either the baseline or modified regression models, suggesting they are properly specified and include relevant variables in their functional form.

### **3.2: F-Tests:**

As the model makes use of 4 separate sophisticated regressions, it is important to make comparisons. Joint significance F-tests are useful for comparing the baseline regressions and modified regressions because they assess whether the included variables collectively contribute explanatory power to the model.

## **4. Results:**

**Table. 5: Baseline Regressions**

| <b>Regressions:</b>                  | <b>(i)</b>             | <b>(ii)</b>          |
|--------------------------------------|------------------------|----------------------|
| <b>Dependent Variables (Models):</b> | <b>net_returns</b>     | <b>gross_returns</b> |
| <b>Independent Variables</b>         |                        |                      |
| <i>Constant</i>                      | -6.1158*<br>[-20.76]   | -6.0428*<br>[-20.14] |
| <i>B_AVG</i>                         | 0.4294<br>[1.29]       | 0.4203<br>[1.25]     |
| <i>AVG</i>                           | 0.9184*<br>[2.97]      | 0.9048*<br>[2.88]    |
| <i>A_AVG</i>                         | 1.1128*<br>[3.41]      | 1.0759*<br>[3.25]    |
| <i>HIGH</i>                          | 2.5196*<br>[6.39]      | 2.5089*<br>[6.29]    |
| <i>Size</i>                          | -0.00007<br>[-0.97]    | -0.00008<br>[-1.05]  |
| <i>fund age</i>                      | -0.0110 ***<br>[-1.71] | -0.0099<br>[-1.52]   |
| <i>No. of observations</i>           | 899                    | 891                  |
| <i>R<sup>2</sup></i>                 | 0.0578                 | 0.0564               |

t-statistics values are depicted where \* represents 1% significance level, \*\*=5% significance level, and \*\*\*=10% significance level.

### **4.1.1: Statistical Significance:**

- Most independent variables in both baseline regressions are significant at 1% level, except fund age at 10%. Below average sustainability ratings do not demonstrate statistical significance.
- Dummy variables for Average, Above average, and High sustainability ratings have significant coefficients with p-values < 0.05. This means funds with these ratings statistically outperform "Low" funds.
- Both baseline regressions have significant constant terms with p-values < 1%, showing that the baseline return is consistently distinct from zero.

### **4.1.2: Economic Significance:**

- The consistent values for Low indicate the dummy variable when all other ratings are zero, thus its coefficients may be categorised. In the baseline regression, the coefficient is negative and 6.1158, indicating that net returns decline by 6.1158% for every unit rise in Low sustainability ratings.
- Compared to the reference group (Low), below-average sustainability ratings improve net returns by -5.6864% per unit. The Average sustainability rating increased by -5.1974%, while the above average and High ratings increased by -5.003% and -3.5962%. Even though these values are negative, High sustainability ratings are higher than low values, indicating that as ratings increase by category from

Low to High, Net returns values tend to be positive and higher, indicating that sustainability ratings show significant positive values to the regression.

- Gross returns have comparable coefficients to net returns, such as 2.5089 for High vs 2.5196 for regression using net returns.
- According to the coefficients, for every unit increase in High value of sustainability ratings, gross returns increase by -3.5339%, while net returns (-3.5962) are much lower. After deducting expenses, net returns should be lower than gross returns. Yes, in ideal firm situations.

**Table. 6: Our regressions**

| <b>Regressions:</b>                          | <b>(i)</b>           | <b>(ii)</b>          | <b>(iii)</b>         | <b>(iv)</b>          |
|----------------------------------------------|----------------------|----------------------|----------------------|----------------------|
| <b>Dependent Variables (Models):</b>         | <b>net_returns</b>   | <b>net_returns</b>   | <b>gross_returns</b> | <b>gross_returns</b> |
| <b>Independent Variables</b>                 |                      |                      |                      |                      |
| <i>Constant</i>                              | -5.8607*<br>[-15.31] | -5.8545*<br>[-15.29] | -5.7295*<br>[-14.69] | -5.7243*<br>[-14.67] |
| <i>B_AVG</i>                                 | 0.4351<br>[1.31]     | 0.4359<br>[1.31]     | 0.4251<br>[1.26]     | 0.4259<br>[1.27]     |
| <i>AVG</i>                                   | 0.9177*<br>[2.97]    | 0.9191*<br>[2.98]    | 0.9037*<br>[2.88]    | 0.9051*<br>[2.89]    |
| <i>A_AVG</i>                                 | 1.1155*<br>[3.42]    | 1.116*<br>[3.43]     | 1.0766*<br>[3.25]    | 1.0773*<br>[3.25]    |
| <i>HIGH</i>                                  | 2.5346*<br>[6.42]    | 2.5336*<br>[6.41]    | 2.5266*<br>[6.32]    | 2.5258*<br>[6.32]    |
| <i>Winsorized log (size)</i>                 | -0.0627<br>[-1.10]   | -0.0615<br>[-1.08]   | -0.0762<br>[-1.32]   | -0.0752<br>[-1.30]   |
| <i>Winsorized fund age [one percentile]</i>  | -0.0098<br>[-1.44]   | -                    | -0.0082<br>[-1.20]   | -                    |
| <i>Winsorized fund age [five percentile]</i> | -                    | -0.0108<br>[-1.48]   | -                    | -0.0091<br>[1.23]    |
| <i>No. of observations</i>                   | 899                  | 899                  | 891                  | 891                  |
| <i>R<sup>2</sup></i>                         | 0.0582               | 0.0583               | 0.0572               | 0.0573               |

t-statistics values are depicted in square brackets, where \* represents the 1% significance level.

#### **4.2.1: Statistical Significance:**

- Most of the variables that are statistically significant [Table. 6] in baseline regressions are also significant in the 4 regressions modified by us.
- It is observed that many dummy variables are significant at 1% level except B\_AVG remains statistically insignificant, reinforcing their minimal impact on returns.
- The winsorized log of size regressor is winsorized at one percent cutoff and is not statistically significant; however, the magnitude of this coefficient is more prominent as compared to the baseline regression of size where the coefficients were too small and closer to zero.
- The winsorized fund age regressor is depicted as two separate regressors (1 percentile and 5 percentile).

#### **4.2.2: Economic Significance:**

- For every unit increase in low value ratings in the regression [Table. 6] the net returns decreases by 5.9%

and it decreases by 5.7% points for gross returns. The magnitude of the constant term in winsorized models is consistent with the baseline models, reinforcing its role as a reliable benchmark for returns.

- For an unit increase in average rating the gross returns reduces by 4.8%, and it decreases net returns by 4.9%.
- With respect to an unit increase in High value of the multiple regression, there is a decrease in 3.2% while the net returns decreases by 3.3%.
- On an average for one percent increase in winsorized log of size there is an approximate decrease in 7.6% of gross returns, and a decrease of approximately 6.2% in net returns.
- Winsorized models of fund age at one percent cutoff shows that for one percent increase there is decrease of 0.9% of net returns in model (i) as compared to 0.8% decrease in gross returns from model (ii). Whilst the cutoff for 5 percent shows there is decrease in 1% of net returns in model (iii) and decrease in 0.9% in gross returns in model (iv).

It is observed that making transformations like log, and winsorizations for fund size and fund age has given a minor increase in  $R^2$ , which seems like a better fit than the baseline model.

#### 4.3: F-tests [Joint Significance tests]:

H0: The coefficients of certain variables are jointly zero ( $\beta_1=\beta_2=\beta_3=0$ )

H1: The coefficients of at least one coefficients is not equal to zero.

**Table. 7: Joint significance test results for baseline and our regressions**

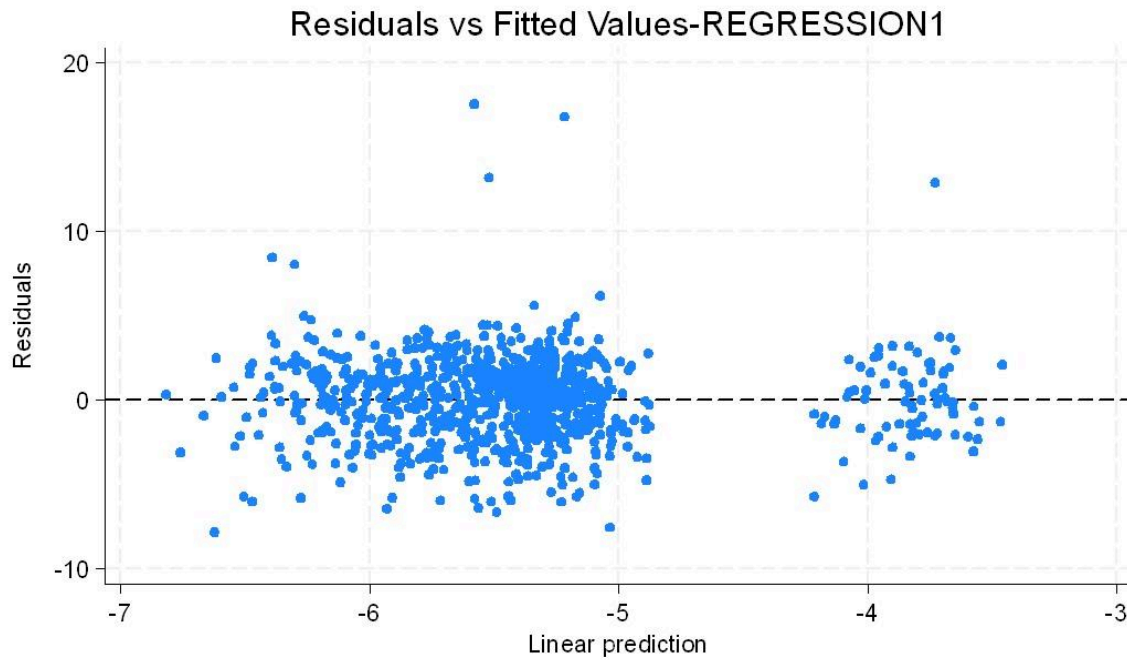
| No. | Test: Joint F-Test                 | Variables tested                                             | Test statistic | p-value | Conclusion                        |
|-----|------------------------------------|--------------------------------------------------------------|----------------|---------|-----------------------------------|
| 1.  | Baseline regression: net returns   | B_AVG, AVG, A-AVG, HIGH, Size and fund age                   | F=9.11         | p=0.00  | Variables are jointly significant |
| 2.  | Baseline regression: gross returns | B_AVG, AVG, A-AVG, HIGH, Size and fund age                   | F=8.81         | p=0.00  | Variables are jointly significant |
| 3.  | Our regression-i                   | B_AVG, AVG, A-AVG, HIGH, Wins_ln_Size and Wins_fund_age one  | F=9.18         | p=0.00  | Variables are jointly significant |
| 4.  | Our regression-ii                  | B_AVG, AVG, A-AVG, HIGH, Wins_ln_Size and Wins_fund_age one  | F=9.20         | p=0.00  | Variables are jointly significant |
| 5.  | Our regression-iii                 | B_AVG, AVG, A-AVG, HIGH, Wins_ln_Size and Wins_fund_age five | F=8.94         | p=0.00  | Variables are jointly significant |
| 6.  | Our regression-iv                  | B_AVG, AVG, A-AVG, HIGH, Wins_ln_Size and Wins_fund_age five | F=8.96         | p=0.00  | Variables are jointly significant |

Our estimated 4 regressional model's [Table. 7] p-values are less than 0.05, indicating that at least one regressor affects the dependent variable. Our factors have a statistically significant effect on gross/net returns. Hence, we reject the null hypothesis that the coefficients of all explanatory variables are jointly equal to zero and these

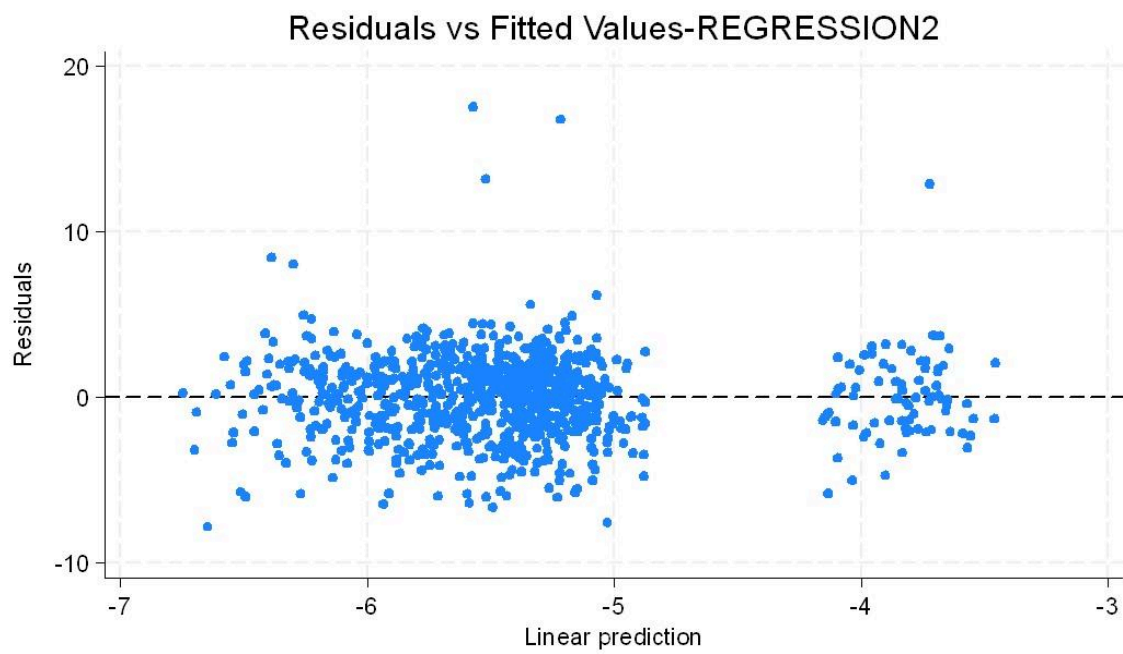
variables in the group improve the model's ability to explain the dependent variable [gross/net returns].

#### **4.4: Homoskedasticity vs Heteroskedasticity:**

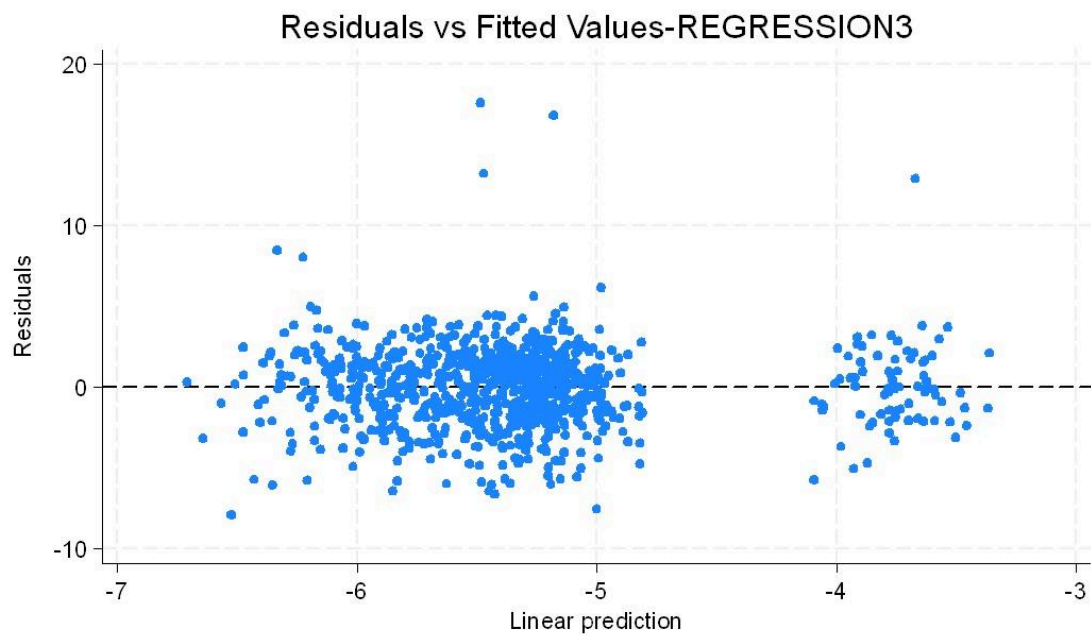
To understand the residual values of errors against the fitted values, we performed visual inspections (Fig. 11-14).



**Fig. 11: Scatter plot for Regression-i**

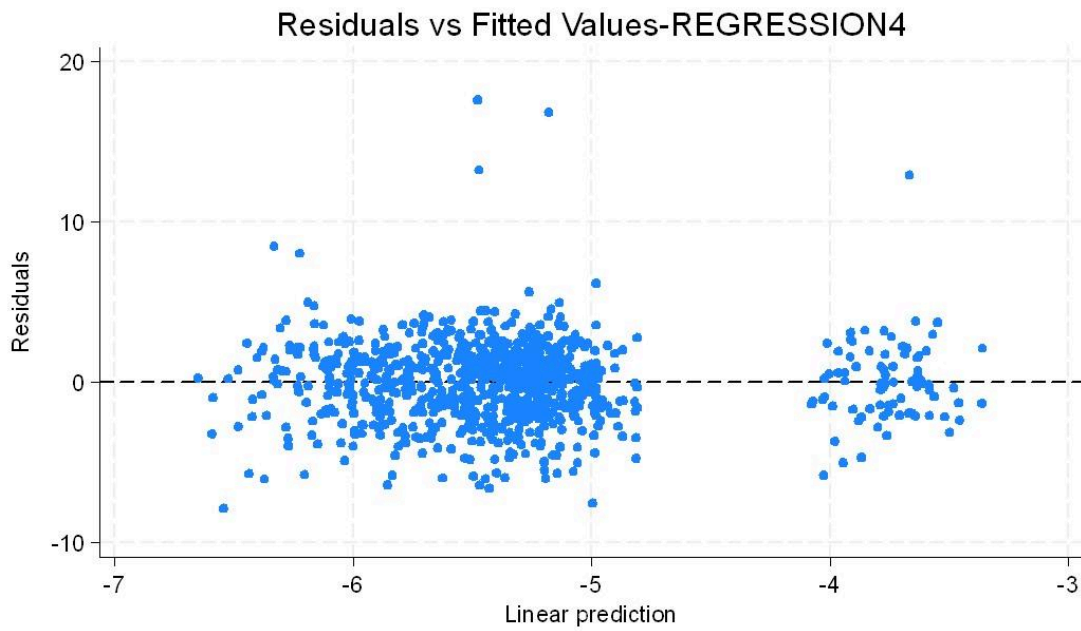


**Fig. 12: Scatter plot for Regression-ii**



**Fig. 13: Scatter plot for Regression-iii**





**Fig. 14: Scatter plot for Regression-iv**

We see residual values spread around the mean, indicating homoskedasticity in our samples. However, we do a white test to verify our results.

H0: Homoskedasticity is present.

H1: Heteroskedasticity is present.

**Table. 8: White Test: Testing for Heteroskedasticity**

| No. | Test: White Test                          | p-value  | Conclusion                  |
|-----|-------------------------------------------|----------|-----------------------------|
| 1.  | <i>Baseline regression: net_returns</i>   | p=0.9668 | Homoskedasticity is present |
| 2.  | <i>Baseline regression: gross_returns</i> | p=0.9608 | Homoskedasticity is present |
| 3.  | <i>Our regression-i</i>                   | p=0.4434 | Homoskedasticity is present |
| 4.  | <i>Our regression-ii</i>                  | p=0.4183 | Homoskedasticity is present |
| 5.  | <i>Our regression-iii</i>                 | p=0.4561 | Homoskedasticity is present |
| 6.  | <i>Our regression-iv</i>                  | p=0.4296 | Homoskedasticity is present |

- No significant heteroskedasticity [Table. 8] was detected in the winsorized models along with the baseline model, suggesting the reliability of the reported standard errors.

## **5. Discussions:**

- The standard errors for most regressors are trustworthy, but B\_AVG, wins\_ln\_size, and wins\_fund\_age\_one are larger than their coefficients, which may indicate significant uncertainty and inaccurate estimations. AVG, A\_AVG, and HIGH have moderate to small standard errors, especially HIGH, indicating better confidence in these values. Sample size and weak regressors may affect coefficient estimate precision, as seen by standard error's size.
- The computed regressions do not exhibit multicollinearity due to the non-evidence of large  $R^2$  and standard errors; additionally, there is no trace of pairwise correlation among regressors.
- It is noticeable that the population correlation coefficients are relatively low; after our estimated model, we see a minute increase; however, these estimates are approximately equal to the baseline model. This could be because of issues with the given dataset; the best solution is to increase sample size or add more regressors to the equation; additionally, it is fairly common to see very low  $R^2$  in cross-sectional data like ours.

### **5.1: Advantages of our model:**

- Multiple regression analysis facilitates ceteris paribus analysis by enabling explicit control over various factors, such as fund size and fund age, that concurrently influence the dependent variables (net\_return and gross\_return).
- Omitted variables are less likely to be a problem in the regression.
- Taking the log of variables often narrows its range; population variables tend to vary widely.
- Though we use a level-log model, the interpretations of this regression, especially with the size regressor, make it look like a constant elasticity model.

### **5.2: Disadvantages of our model:**

- Sometimes the log of variables between 0 and 1, but closer to 0, the log version of such variables could be negative and larger in magnitude, which could cause us to give wrong estimates of coefficients.
- Our models are complicated because there are 4 regressions; we must evaluate them with many tests for stating comparisons between them.

## **6. Conclusions:**

- Our analysis of fund performance indicates that most companies exhibit negative returns, whether net or gross. It is evident that a higher sustainability rating, such as an above-average rating, correlates with a greater slope compared to below-average ratings, suggesting that higher sustainability ratings are associated with minimised negative returns. Hence the sustainability rating does affect net and gross returns, bringing less negative returns when compared to a rating value below it in the regression equation.
- The firm's size and age have generally exhibited an insignificant statistical impact, while fund age has generally shown a negative relationship.
- Higher sustainability ratings are associated with larger gross and net returns. According to the coefficient corresponding to different levels, it can be found that as the level decreases, the coefficient is also constantly decreasing (taking net return as an example:  $2.535 > 1.116 > 0.918 > 0.435$ ), which shows that higher sustainability is related to larger gross\_return and net\_return. This shows that companies with high sustainability have stronger positive impact on gross\_return and net\_return, with only the uncertainty in predicting the below-average value because its estimates may be inaccurate because of its standard errors.

### **6.1: Suggestions**

- Prioritise investment in high-grade funds because they have a stronger positive impact on returns, and for some low-rated funds, optimised operation management could be tried, so it could improve sustainability ratings and avoid the return disadvantages brought about by low ratings.
- Focus on mid-level funds: Enterprises can establish a portfolio monitoring system and regularly evaluate the relationship between fund returns and sustainability ratings to avoid a decrease in high-rated fund returns and can also pay more attention to medium-rated funds.
- Social impact: the competitiveness and social responsibility of high-rated funds in the market can be emphasised so as to attract more customers who pay attention to ESG investment; of course, enterprises can also highlight the high-rated fund and the financial return advantages it brings through brand promotion.

## **7. Limitations:**

1. The model seems to be based on a small group of predictions, which means it might be missing some important factors that affect net returns, like outside economic factors or changes in the market.
2.  $R^2$  is a measure of the goodness of fit of the model, indicating how well the independent variables explain the variability in the dependent variable. Specifically, the outcome about regression shows that  $R^2$  equals 0.0581, which is very low. And it means that 5.81% of the variation in the dependent variable is explained by the independent variables in the model, which may suggest insufficient explanatory power, excessive noise, or issues due to data limitations.
3. There are many other functional possibilities for our regression equation that may improve the model, which we may not have identified.
4. Increasing the bandwidth of the outliers range would sometimes be better than our existing models. However, in this case, there is a risk that some of the data between the outliers may not be included in the sample, resulting in insufficient data within the winsorized variables and thus giving us inaccurate estimates.

## *Appendix*

### \*SUMMARY STATISTICS OF THE GIVEN DATA

summarize net\_return gross\_return sustainabilityrating size fund\_age

\*Creating Dummy Variables for Sustainability Ratings. The Dummy variables we have created are B\_AVG, AVG, A\_AVG AND HIGH, which signifies Below Average, Average, Above Average and High rating values of Sustainability Ratings given in the dataset.

\*We have created 4 dummy variables because including 5 variables causes to the issue of dummy variable trap.

\*The reference dummy is "Low" value of Sustainability Rating which is not included in the regression.

generate LOW = 1 if sustainabilityrating == "Low"

replace LOW = 0 if LOW == .

generate B\_AVG = 1 if sustainabilityrating == "Below Average"

replace B\_AVG = 0 if B\_AVG == .

generate AVG = 1 if sustainabilityrating == "Average"

replace AVG = 0 if AVG == .

generate A\_AVG = 1 if sustainabilityrating == "Above Average"

replace A\_AVG = 0 if A\_AVG == .

generate HIGH = 1 if sustainabilityrating == "High"

replace HIGH = 0 if HIGH == .

### \*SUMMARY STATISTICS OF THE GIVEN DATA INCLUDING THE DUMMY VARIABLES OF SUSTAINABILITY RATINGS

summarize net\_return gross\_return B\_AVG AVG A\_AVG HIGH size fund\_age

\*are outliers present in size??, answer-yes they are found

graph box size

generate ln\_size=ln(size)

\*are outliers present in ln\_size

graph box ln\_size

\*are outliers present in fund\_age??

graph box fund\_age

\*mitigating outliers in ln\_size

winsor ln\_size , p(0.1) gen(wins\_ln\_size)

\*checking outliers for wins\_ln\_size

graph box wins\_ln\_size

\*mitigating outliers in ln\_size in 5% values

winsor ln\_size , p(0.05) gen(wins\_ln\_size\_)

\*checking outliers for wins\_ln\_size\_ which is for lower than 5%ile and higher than 95%ile-using this is a better option I guess

graph box wins\_ln\_size\_

\*size is too large pre-log tranformation

tabstat size fund\_age, stat(mean sd min max p1 p5 median p75 p99)

\*post log transformation

tabstat size ln\_size, stat(mean sd min max p1 p5 median p75 p99)

graph box ln\_size

\*post winsorization of ln\_size

graph box wins\_ln\_size\_

\*mitigating outliers in fund\_age at 1 percentile

winsor fund\_age , p(0.01) gen(wins\_fund\_age\_one)

graph box wins\_fund\_age\_one

\*mitigating outliers in fund\_age at 5 percentile

winsor fund\_age , p(0.05) gen(wins\_fund\_age\_five)

graph box wins\_fund\_age\_five

summarize net\_return gross\_return B\_AVG AVG A\_AVG HIGH wins\_ln\_size\_\_ fund\_age wins\_fund\_age\_one  
wins\_fund\_age\_five

\*we are not using winsorized values of fundage of 10% and 25% because it is not necessary to bring the values of fundage similar to the range of the other control variables used in the regression, however we make two regression equations with winzorized values of fundage one which has been winzorised to 1 percent and the other regression is winzorised to 5 percent of lower and upper values.

reg net\_return B\_AVG AVG A\_AVG HIGH wins\_ln\_size\_ wins\_fund\_age\_one

reg gross\_return B\_AVG AVG A\_AVG HIGH wins\_ln\_size\_ wins\_fund\_age\_one

reg net\_return B\_AVG AVG A\_AVG HIGH wins\_ln\_size\_ wins\_fund\_age\_five

reg gross\_return B\_AVG AVG A\_AVG HIGH wins\_ln\_size\_ wins\_fund\_age\_five

\*Baseline regression

reg net\_return B\_AVG AVG A\_AVG HIGH size fund\_age

reg gross\_return B\_AVG AVG A\_AVG HIGH size fund\_age

\*Testing for heteroskedasticity

\*At first we must conduct f tests and check for joint significance tests with the independent variables, first we perform with baseline regression and then with our identified regression.

\*Baseline regression1

reg net\_return B\_AVG AVG A\_AVG HIGH size fund\_age

test B\_AVG=AVG

test AVG=A\_AVG

test A\_AVG=HIGH

test size=fund\_age

\*checking for joint significance of the independent variables in baseline regression

test B\_AVG AVG A\_AVG HIGH size fund\_age

\*Baseline regression2

reg gross\_return B\_AVG AVG A\_AVG HIGH size fund\_age

test B\_AVG=AVG

test AVG=A\_AVG

test A\_AVG=HIGH

test size=fund\_age

\*checking for joint significance of the independent variables in baseline regression

test B\_AVG AVG A\_AVG HIGH size fund\_age

\*Our regression\_1:

reg net\_return B\_AVG AVG A\_AVG HIGH wins\_ln\_size\_wins\_fund\_age\_one

test B\_AVG=AVG

test AVG=A\_AVG

test A\_AVG=HIGH

test wins\_ln\_size\_wins\_fund\_age\_one

\*checking for joint significance of the independent variables in our regression

test B\_AVG AVG A\_AVG HIGH wins\_ln\_size\_wins\_fund\_age\_one

\*Our regression\_2:

reg net\_return B\_AVG AVG A\_AVG HIGH wins\_ln\_size\_wins\_fund\_age\_five

test B\_AVG=AVG

test AVG=A\_AVG

test A\_AVG=HIGH

test wins\_ln\_size\_wins\_fund\_age\_five

\*checking for joint significance of the independent variables in our regression

test B\_AVG AVG A\_AVG HIGH wins\_ln\_size\_wins\_fund\_age\_five

\*Our regression\_3:

reg gross\_return B\_AVG AVG A\_AVG HIGH wins\_ln\_size\_wins\_fund\_age\_one

test B\_AVG=AVG

test AVG=A\_AVG

test A\_AVG=HIGH

test wins\_ln\_size\_wins\_fund\_age\_one

\*checking for joint significance of the independent variables in our regression

test B\_AVG AVG A\_AVG HIGH wins\_ln\_size\_ wins\_fund\_age\_one

\*Our regression\_4:

reg gross\_return B\_AVG AVG A\_AVG HIGH wins\_ln\_size\_ wins\_fund\_age\_five

test B\_AVG=AVG

test AVG=A\_AVG

test A\_AVG=HIGH

test wins\_ln\_size\_=wins\_fund\_age\_five

\*checking for joint significance of the independent variables in our regression

test B\_AVG AVG A\_AVG HIGH wins\_ln\_size\_ wins\_fund\_age\_five

\*All of the above regressions are jointly significant, indicating the coefficients are statistically different from zero, and thus we reject the null hypothesis, stating that the H1 is true and that there is presence of heteroskedasticity, we have rejected H0 and thus homoskedasticity is not present. We proceed to white test.

\*WHITE TEST FOR BASELINE REGRESSION1

reg net\_return B\_AVG AVG A\_AVG HIGH size fund\_age

\*checking for joint significance of the independent variables in baseline regression

test B\_AVG AVG A\_AVG HIGH size fund\_age

estat imtest, white

\*the p value is 0.9668, which is much higher than 0.05, and so there is no evidence that it is heteroskedastic, and hence we fail to reject the null hypothesis of homoskedasticity, which implies that this regression is not heteroskedastic and thus it is homoskedastic, where the null hypothesis is true. So, hence we do not perform robust standard errors into the regression

\*WHITE TEST FOR BASELINE REGRESSION2

reg gross\_return B\_AVG AVG A\_AVG HIGH size fund\_age

\*checking for joint significance of the independent variables in baseline regression

test B\_AVG AVG A\_AVG HIGH size fund\_age

estat imtest, white

\*the p value is 0.9608, which is much higher than 0.05, and so there is no evidence that it is heteroskedastic, and hence we fail to reject the null hypothesis of homoskedasticity, which implies that this regression is not heteroskedastic and thus it is homoskedastic, where the null hypothesis is true. So, hence we do not perform robust standard errors into the regression

\*WHITE TEST FOR OUR REGRESSION 1

reg net\_return B\_AVG AVG A\_AVG HIGH wins\_ln\_size\_ wins\_fund\_age\_one

estat imtest, white

\*the p value is 0.4434, which is much higher than 0.05, and so there is no evidence that it is heteroskedastic, and hence we fail to reject the null hypothesis of homoskedasticity, which implies that this regression is not heteroskedastic and thus it is homoskedastic, where the null hypothesis is true. So, hence we do not perform robust standard errors into the regression1

#### \*WHITE TEST FOR OUR REGRESSION 2

```
reg net_return B_AVG AVG A_AVG HIGH wins_ln_size_ wins_fund_age_five
estat imtest, white
```

\*the p value is 0.4183, which is much higher than 0.05, and so there is no evidence that it is heteroskedastic, and hence we fail to reject the null hypothesis of homoskedasticity, which implies that this regression is not heteroskedastic and thus it is homoskedastic, where the null hypothesis is true. So, hence we do not perform robust standard errors into the regression1

#### \*WHITE TEST FOR OUR REGRESSION 3

```
reg gross_return B_AVG AVG A_AVG HIGH wins_ln_size_ wins_fund_age_one
estat imtest, white
```

\*the p value is 0.4561, which is much higher than 0.05, and so there is no evidence that it is heteroskedastic, and hence we fail to reject the null hypothesis of homoskedasticity, which implies that this regression is not heteroskedastic and thus it is homoskedastic, where the null hypothesis is true. So, hence we do not perform robust standard errors into the regression1

#### \*WHITE TEST FOR OUR REGRESSION 4

```
reg gross_return B_AVG AVG A_AVG HIGH wins_ln_size_ wins_fund_age_five
estat imtest, white
```

\*the p value is 0.4296, which is much higher than 0.05, and so there is no evidence that it is heteroskedastic, and hence we fail to reject the null hypothesis of homoskedasticity, which implies that this regression is not heteroskedastic and thus it is homoskedastic, where the null hypothesis is true. So, hence we do not perform robust standard errors into the regression1

#### \*RESIDUAL PLOTS FOR OUR REGRESSIONS:

##### \*REGRESSION 1

```
reg net_return B_AVG AVG A_AVG HIGH wins_ln_size_ wins_fund_age_one
predict fitted_values10, xb // Generate fitted values (estimates of y)
predict residuals10, residuals // Generate residuals (y - fitted values)
scatter residuals10 fitted_values10, yline(0) ///
    xlabel(, grid) ylabel(, grid) ///
    title("Residuals vs Fitted Values-REGRESSION1")
```

##### \*REGRESSION 2

```
reg net_return B_AVG AVG A_AVG HIGH wins_ln_size_ wins_fund_age_five
predict fitted_values8, xb // Generate fitted values (estimates of y)
predict residuals8, residuals // Generate residuals (y - fitted values)
scatter residuals8 fitted_values8, yline(0) ///
    xlabel(, grid) ylabel(, grid) ///
    title("Residuals vs Fitted Values-REGRESSION2")
```

##### \*REGRESSION 3



```

reg gross_return B_AVG AVG A_AVG HIGH wins_ln_size_ wins_fund_age_one
predict fitted_values7, xb // Generate fitted values (estimates of y)
predict residuals7, residuals // Generate residuals (y - fitted values)
scatter residuals7 fitted_values7, yline(0) ///
    xlabel(, grid) ylabel(, grid) ///
    title("Residuals vs Fitted Values-REGRESSION3")

```

\*REGRESSION4

```

reg gross_return B_AVG AVG A_AVG HIGH wins_ln_size_ wins_fund_age_five
predict fitted_values9, xb // Generate fitted values (estimates of y)
predict residuals9, residuals // Generate residuals (y - fitted values)
scatter residuals9 fitted_values9, yline(0) ///
    xlabel(, grid) ylabel(, grid) ///
    title("Residuals vs Fitted Values-REGRESSION4")

```

\*Performing Ramseys reset test to check for functional form, H0: the model is correctly specified, H1: the model is incorrectly specified

\*Performing it on the baseline regressions

```

reg net_return B_AVG AVG A_AVG HIGH size fund_age
estat ovtest

```

\*The p value is 0.4023>0.05 hence we fail to reject null hypothesis, thus the model is correctly specified

```

reg gross_return B_AVG AVG A_AVG HIGH size fund_age
estat ovtest

```

\*The p value is 0.3670>0.05 hence we fail to reject null hypothesis, thus the model is correctly specified

\*Performing Ramseys reset test to check for wrong functional form, H0: the model is correctly specified, H1: the model is incorrectly specified

\*Performing it on our regression1

```

reg net_return B_AVG AVG A_AVG HIGH wins_ln_size_ wins_fund_age_one
estat ovtest

```

\*The p value is 0.4133>0.05 hence we fail to reject null hypothesis, thus the model is correctly specified

\*Performing it on our regression2

```

reg net_return B_AVG AVG A_AVG HIGH wins_ln_size_ wins_fund_age_five
estat ovtest

```

\*The p value is 0.3998>0.05 hence we fail to reject null hypothesis, thus the model is correctly specified

\*Performing it on our regression3

```

reg gross_return B_AVG AVG A_AVG HIGH wins_ln_size_ wins_fund_age_one
estat ovtest

```

\*The p value is 0.3779>0.05 hence we fail to reject null hypothesis, thus the model is correctly specified

\*Performing it on our regression4

```

reg gross_return B_AVG AVG A_AVG HIGH wins_ln_size_ wins_fund_age_five
estat ovtest

```

\*The p value is 0.3529>0.05 hence we fail to reject null hypothesis, thus the model is correctly specified

\*Hence we can state that our regressions are correct with their functional forms

## References

Brooks, C. (2019). *Introductory Econometrics for Finance*. 4th ed. Cambridge, United Kingdom ; New York, Ny: Cambridge University Press.

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